

Pythagoras' theorem

Pythagoras sats

MATH

Discovered 1800 BC · Babylon

EUREKA

RARE

01 / 10

Percent change

Förändringsfaktor

MATH

Discovered 14 · Roman taxes

EUREKA

COMMON

02 / 10

Newton's second law

Newtons andra lag

PHYSICS

Discovered 1687 · Isaac Newton

EUREKA

LEGENDARY

03 / 10

Energy conservation

Energiprincipen

PHYSICS

Discovered 1847 · Mayer and Joule

EUREKA

RARE

04 / 10

Ohm's law

Ohms lag

PHYSICS

Discovered 1827 · Georg Ohm

EUREKA

COMMON

05 / 10

Density

Densitet

CHEMISTRY

Discovered 250 BC · Archimedes

EUREKA

COMMON

06 / 10

Oxygen

Syre

CHEMISTRY

Discovered 1772 · C. W. Scheele

EUREKA

COMMON

07 / 10

Gold

Guld

CHEMISTRY

Discovered 4500 BC · Varna, Bulgaria

EUREKA

LEGENDARY

08 / 10

DNA

DNA

BIOLOGY

Discovered 1953 · Franklin, Watson, Crick

EUREKA

LEGENDARY

09 / 10

Newton's second law



THE IDEA

A total force changes how something moves, and the heavier it is, the less it changes.

$$F = m \cdot a$$

LETS YOU PREDICT

Knowing force and mass, where a thing will be later.

TESTS USUALLY ASK

- Find the acceleration
- Braking or friction force



Scan for the full explanation and a practice question.

EUREKA · 03 / 10

Percent change



THE IDEA

A change is measured against where you started, so you multiply by a change factor.

$$\text{new} = \text{old} \times \text{factor}$$

LETS YOU PREDICT

Where a price ends after many changes: old $\times$ factorⁿ.

TESTS USUALLY ASK

- Up 20 %, then down 20 %: back to start?
- Interest over several years



Scan for the full explanation and a practice question.

EUREKA · 02 / 10

Pythagoras' theorem



THE IDEA

In a right-angled triangle, the squares on the two short sides add up to the square on the long side.

$$a^2 + b^2 = c^2$$

LETS YOU PREDICT

A distance you cannot measure, from two you can.

TESTS USUALLY ASK

- Find the missing side
- Is this triangle right-angled?



Scan for the full explanation and a practice question.

EUREKA · 01 / 10

Density



THE IDEA

Density is how much mass is packed into each bit of volume; the less dense thing floats.

$$\rho = m / V$$

LETS YOU PREDICT

Whether something floats before you drop it in.

TESTS USUALLY ASK

- Calculate a density
- Which substance is it?



Scan for the full explanation and a practice question.

EUREKA · 06 / 10

Ohm's law



THE IDEA

The current through a resistor grows in step with the voltage across it.

$$U = R \cdot I$$

LETS YOU PREDICT

The current before you switch on.

TESTS USUALLY ASK

- Find current, voltage or resistance
- Resistors in series



Scan for the full explanation and a practice question.

EUREKA · 05 / 10

Energy conservation



THE IDEA

Energy never appears or disappears; it only changes form, so the total stays the same.

$$m \cdot g \cdot h = \frac{1}{2} \cdot m \cdot v^2$$

LETS YOU PREDICT

Speed at the bottom from the height alone.

TESTS USUALLY ASK

- Speed of a falling object
- Where did the energy go?



Scan for the full explanation and a practice question.

EUREKA · 04 / 10

DNA



THE IDEA

DNA is a code of four letters on a twisted ladder, and each half tells you the other.

$$\text{A pairs with T} \cdot \text{C pairs with G}$$

LETS YOU PREDICT

One strand gives you the other strand.

TESTS USUALLY ASK

- Write the partner strand
- How a cell copies DNA



Scan for the full explanation and a practice question.

EUREKA · 09 / 10

Gold



THE IDEA

Gold is a noble metal: it hardly reacts, so it never rusts and stays shiny for thousands of years.

$$\text{Au} \cdot \text{atomic number } 79 \cdot \rho = 19.3 \text{ g/cm}^3$$

LETS YOU PREDICT

Treasure found after 3,000 years still shines.

TESTS USUALLY ASK

- Reactivity: why gold does not rust
- Carats as percent



Scan for the full explanation and a practice question.

EUREKA · 08 / 10

Oxygen



THE IDEA

Oxygen has 6 outer electrons and wants 2 more, so it reacts with almost everything.

$$\text{O} \cdot \text{atomic number } 8 \cdot \text{shells } 2, 6$$

LETS YOU PREDICT

It forms O²⁻ ions and two bonds.

TESTS USUALLY ASK

- Electron shells and ions
- Burning = reacting with oxygen

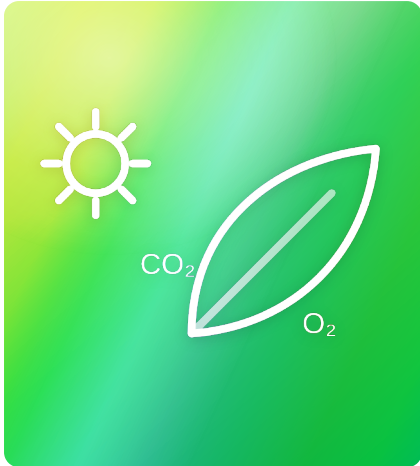


Scan for the full explanation and a practice question.

EUREKA · 07 / 10

Photosynthesis

Fotosyntes



BIOLOGY

RARE

Discovered 1779 · Jan Ingenhousz

EUREKA

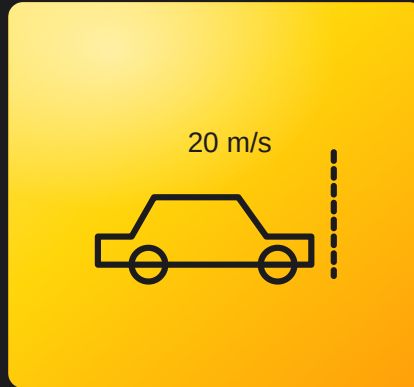
10 / 10

CHALLENGE

LAB DUEL

Stop a car doing 20 m/s within 30 m.

What braking does it take, and is it possible?



Play a card that solves it and say why in one sentence.

EUREKA

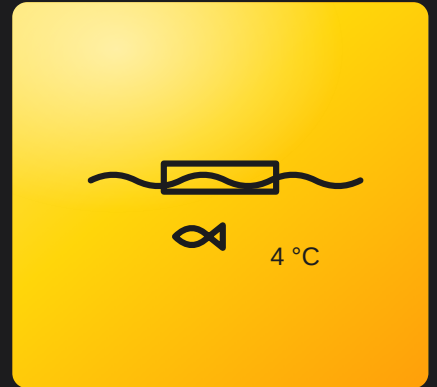
C1

CHALLENGE

LAB DUEL

Why does ice float on a lake?

Explain it so a friend could predict it.



Play a card that solves it and say why in one sentence.

EUREKA

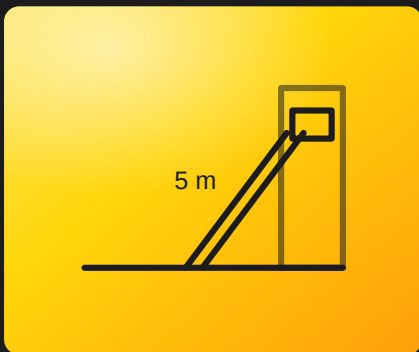
C2

CHALLENGE

LAB DUEL

Will a 5 m ladder reach a window 4 m up if its foot must stand 3 m out?

Yes or no, and prove it.



Play a card that solves it and say why in one sentence.

EUREKA

C3

Accepted answers

C2

Density

Oxygen

Two of these together score double.

A WINNING EXPLANATION

Ice (0.92 g/cm^3) is less dense than water (1.00 g/cm^3), so it floats. Bonus: water molecules, built around oxygen, hold each other further apart when they freeze.

$0.92 < 1.00 \rightarrow$ floats, about 9/10 under water

Close enough counts: the other players decide if the explanation is right.

Accepted answers

C1

Newton's second law

Energy conservation

Two of these together score double.

A WINNING EXPLANATION

The car's movement energy $\frac{1}{2}mv^2$ must be removed by the braking force over 30 m, so it needs about 6.7 m/s^2 of braking: $F = m \cdot a$, about 8 kN for a 1,200 kg car. Possible on a dry road if the brakes go on at once, not on ice.

$$a = v^2 / (2s) = 400 / 60 \approx 6.7 \text{ m/s}^2$$

Close enough counts: the other players decide if the explanation is right.

Photosynthesis



THE IDEA

Plants use light to build sugar from carbon dioxide and water, and give off oxygen.



LETS YOU PREDICT

Most of a tree's mass comes from the air.

TESTS USUALLY ASK

- The balanced formula
- Where the energy comes from



Scan for the full explanation and a practice question.

EUREKA · 10 / 10

Accepted answers

C3

Pythagoras' theorem

Only one card fits this one.

A WINNING EXPLANATION

The ladder is the long side: $3^2 + h^2 = 5^2$, so $h^2 = 16$ and $h = 4 \text{ m}$. It reaches exactly to the window sill, with nothing to spare.

$$\sqrt{(25 - 9)} = \sqrt{16} = 4 \text{ m}$$

Close enough counts: the other players decide if the explanation is right.

How to make the prototype deck

1. Print pages 1–4 on thick paper (200–300 g/m²) at **actual size / 100 %**, double-sided, **flip on long edge**. Each back is printed behind its front.
2. No duplex printer? Print page 1, put the sheet back in the tray face down, print page 2. Do one test sheet first and hold it against a window to check the fronts and backs line up.
3. Cut along the card edges. The grid is 3 × 3 cards of 63 × 88 mm with no gap, so every cut is a straight line across the sheet. A 3–4 mm corner punch gives the rounded corners of a real trading card.
4. Laminate the cut cards in glossy pouches (or laminate the whole sheet first, then cut, leaving a small edge).
5. Check the QR codes with a phone. They point to <https://sparkon.cards/eurekav1/cards/...>; if the website address changes, rebuild the sheet with the new address.

What is on this sheet

10 concept cards (Math 2 · Physics 3 · Chemistry 3 · Biology 2) and 3 challenge cards for Lab Duel.

Making the rare ones shine

- **Common**: white frame, printed and laminated in normal gloss.
- **Rare**: silver frame and a rainbow sheen over the picture. For real shine, laminate rare cards in **holographic laminating film** instead of clear.
- **Legendary**: gold frame. For real gold, print on a laser printer and run **toner-reactive gold foil** through a laminator: the foil sticks only where there is toner.
- **π Mystery**: the paper version here is only the prototype. The real card, one in the world, would be made on a **brushed-aluminium card blank** with π in gold foil and its digits laser-engraved round the edge, then hidden in one random booster pack. (Check prices with a metal business-card maker.)

Lab Duel, quick rules for the test rounds

1. Shuffle the concept cards and deal 3 to each player (2–4 players). Put the challenge cards face down in a pile.
2. Turn over a challenge. Anyone may play a card from their hand and explain in one sentence how it solves the challenge. Say a number if you can.
3. Flip the challenge to check. Right card and a sensible explanation: you keep the challenge. Two accepted cards together score double.
4. Played cards go back into your hand. If nobody solves it, the challenge goes to the bottom of the pile.
5. First to 5 points wins. With only 3 challenges in the prototype, play to 3.

Write down during the test rounds

Which challenge caused the most arguing · which card nobody used · any rule question you had to invent an answer for · how long a round took · whether players looked at the card backs to explain.